Home energy, made understandable

The most valuable visual components for a solar, battery and EV homeowner in Woodland Hills.

THE RESEARCH QUESTION

What helps an owner understand what is happening, why it matters and what to do next?



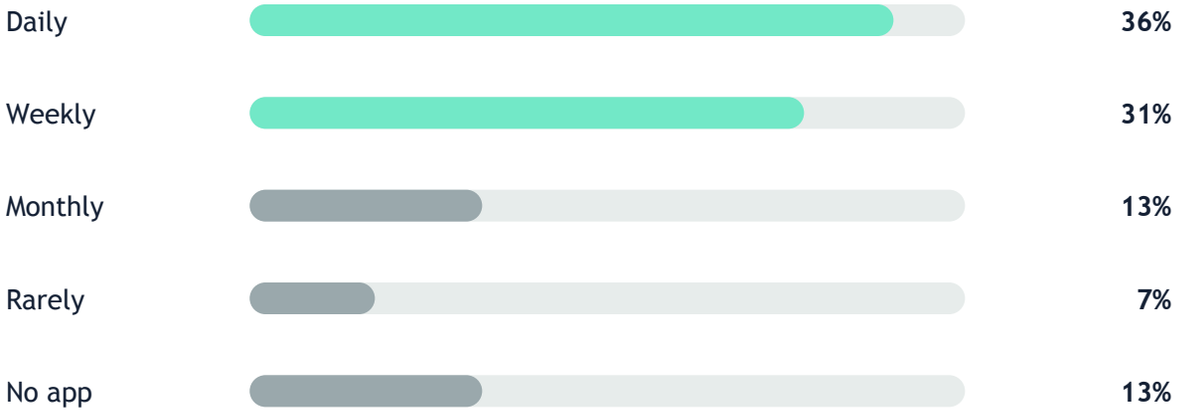
Existing project visual reference - not a researched product screen

Start with what owners ask

Survey and field research are directional. This page separates measured responses from our design inference.

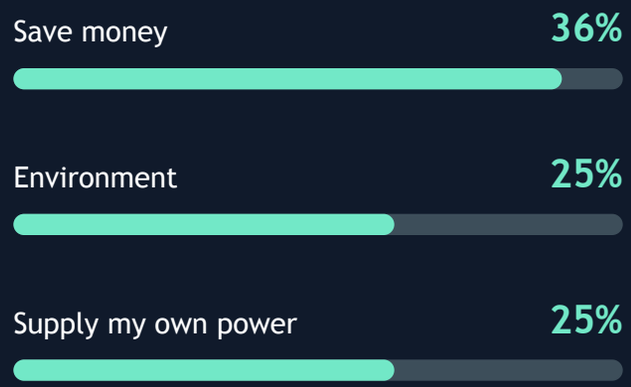
HOW OFTEN SOLAR OWNERS CHECK AN APP

67% daily or weekly in this survey



Source: Enact, 2022 online survey of 482 solar owners (California and New York)

MOTIVATION TO GO SOLAR



Inference: cost clarity belongs beside the energy visual, not buried in a later report.

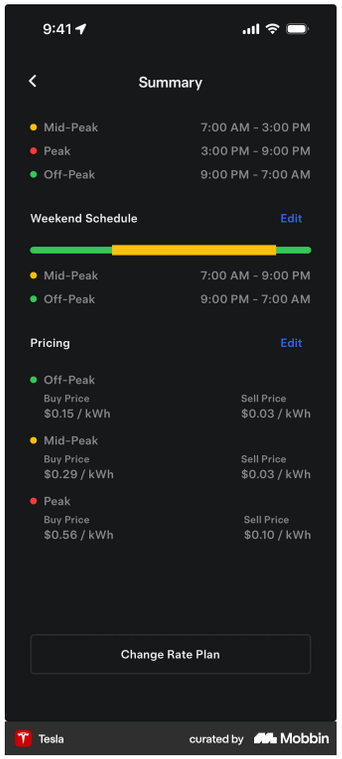
WHAT OTHER EVIDENCE ADDS

A 2026 choice experiment found rate alerts and appliance-level use more valued than carbon feedback in its South Korean sample. Tesla now explains recent battery actions and current prices in its Home Status feature. Together, these point toward explanation and useful action over a wall of numbers.

Zhao and Huh (2026) Tesla Home Status Survey and tariff contexts differ from Woodland Hills.

Patterns worth borrowing

These screens show existing presentation choices. They do not measure which design users prefer.

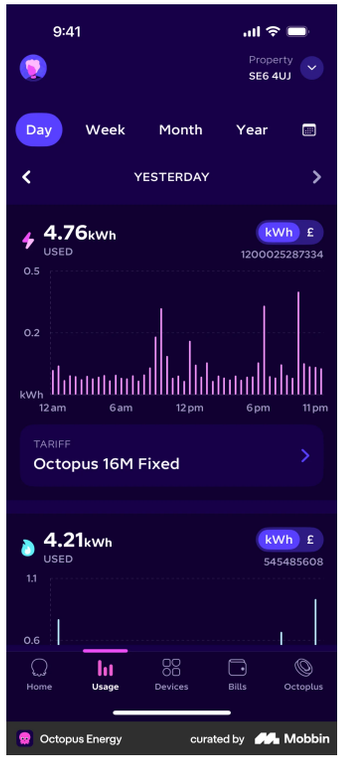


TESLA

Rate periods

When is energy expensive?

View on Mobbin

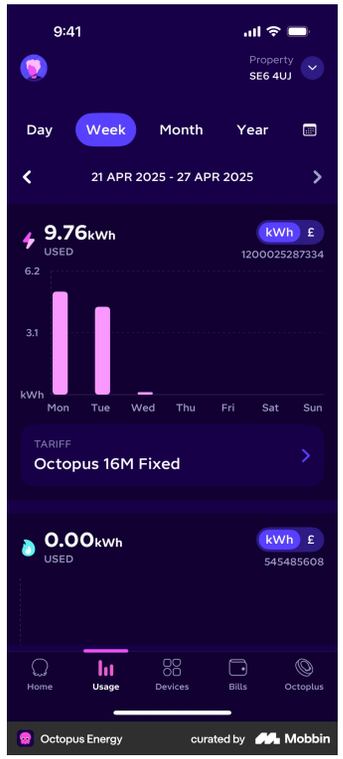


OCTOPUS

Daily use

What happened today?

View on Mobbin

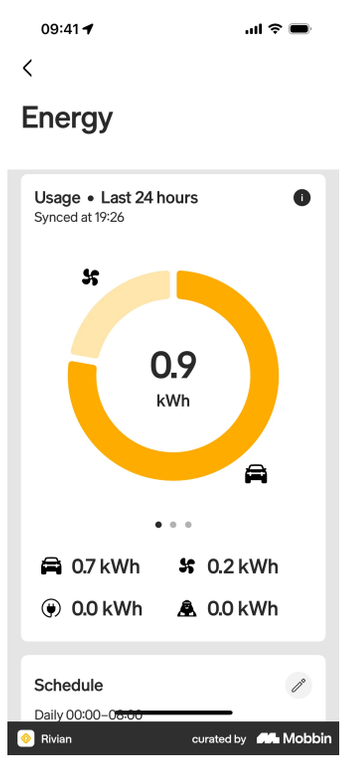


OCTOPUS

Weekly use

What changed this week?

View on Mobbin



RIVIAN

Labeled breakdown

What made up the total?

View on Mobbin

Tesla prices shown are for another utility. Octopus is a UK app. Rivian is vehicle-specific.

Build three components first

Each should answer one homeowner question. Data status refers to the existing simulation.

01 NOW + FLOW

READY

Where is power coming from and going?

Show the five sources/destinations, direction, kW and one factual explanation.

02 COST + CREDIT

PARTIAL

What did this period cost or earn as credit?

Pair the rate period with modeled import cost, export credit and savings.

03 BATTERY READINESS

PARTIAL

What is the battery doing, and what is held in reserve?

Show charge %, direction and reserve - without inventing backup hours.

NEXT: Usage drivers and exact time-window history. SECONDARY: EV schedule and modeled CO2e.

Make the data legible and honest

A beautiful flow is useful only when the homeowner can understand the units, accounting and data limits.

	SAY	DO NOT IMPLY
kW vs kWh	kW now; kWh over time	The units are interchangeable
Export credit	Bill credit earned	Cash paid to homeowner
Savings	Modeled vs all-grid home	Actual LADWP bill total
Battery	Charge and reserve	Guaranteed backup hours

The current project replays simulated 2026 data at one simulated day per minute. Actual bills, live outages, device health and measured appliance use require new data sources.

Key sources: Tesla Energy Data · LADWP rates · LADWP NEM · Enact survey · Zhao and Huh · Full source map in the companion research report.